

FIELD AND NAVAL ARTILLERY

Artillery—guns that are too big and heavy to be fired by hand—include not only cannon but also smaller weapons such as swivel guns. While the design of early artillery used on land or at sea was similar, guns made for ships had to meet special requirements—space aboard ships is limited and the risk of fire considerable. Guns mounted on a pivot—swivel guns—were developed to increase the maneuverability of artillery. Light versions of swivel guns were created for naval use, and these guns could be fit onto sockets on the sides of ships. This helped to stabilize the guns when firing and to absorb recoil. Although most naval guns were muzzle-loading, loading the charge in the breech of the gun's barrel rather than in the muzzle, or breech-loading, made these guns easier to load. This was a useful feature, because it was impractical to reload a muzzle-loader whose muzzle projected from the side of the ship. Field and naval artillery gradually began to use balls of iron and lead rather than stone.

► SWEDISH SWIVEL GUN

Date c.1500

Origin Sweden

Material Iron

Shot Round or grapeshot

Swivel guns first appeared in the late 14th century. Unlike fixed cannon, which could only fire in one direction, they provided an arc of fire, and were mainly breech-loading. This model would have been mounted on a boat or a building and would often be loaded with grapeshot—small balls of iron and lead.



Muzzle lost through corrosion



Forged-iron barrel



FULL VIEW

Match-holder to hold smoldering match-cord

Elaborate sight block holds blades of varying heights for different ranges



Trigger

Lock and furniture made of brass to resist corrosion

Pan holds a quantity of gunpowder, which is ignited by a smoldering match-cord to light the main gunpowder charge in the barrel

Muzzle ring



Hoop iron barrel

▼ ENGLISH HAND-CANNON

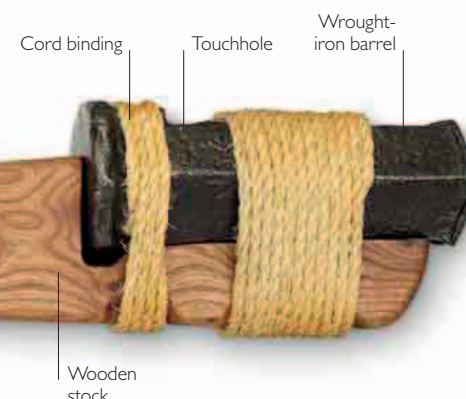
Date 1480

Origin England

Barrel Not known

Caliber Not known

Hand-cannon were really small-scale versions of cannon and were deployed in the same way, but unlike true artillery they were small enough to be carried and fired by one user. Their muzzle-loading barrels were attached to wooden tillers. Small hand-cannon were used in naval and land warfare, but they were difficult to aim. The user had to hold the gun, look where he was aiming, direct the gun using a tiller, and then place a burning match-cord into a small amount of gunpowder around a touchhole—a vent at the rear of the barrel. On ignition, this priming powder would fire the main gunpowder charge in the breech of the barrel.



Modern reproduction of wooden tiller, used to aim the weapon

Wooden stock

Cord binding

Touchhole

Wrought-iron barrel